Team 31: Sbyyaat

GENHACK 2025 · WEEK 2

Urban Heat Island: Madrid Analysis

Visualizing the correlation between temperature anomalies and vegetation density in the Spanish capital.

The Challenge

The Phenomenon

The Urban Heat Island (UHI) effect occurs when urban areas experience significantly warmer temperatures than their rural surroundings due to human activity and surface modification.

Why Madrid?

Madrid presents an ideal case study due to its dense, concrete-heavy urban core contrasted against a dry, semi-arid rural periphery, leading to intense summer heat retention.

Data Pipeline

ERA5-Land

Reanalysis data used to extract 2-meter air temperature (daily maximums) during Summer 2020.

Sentinel-2

High-resolution satellite imagery processed to calculate NDVI (Vegetation Index) for Q3 2020.

GADM & Stations

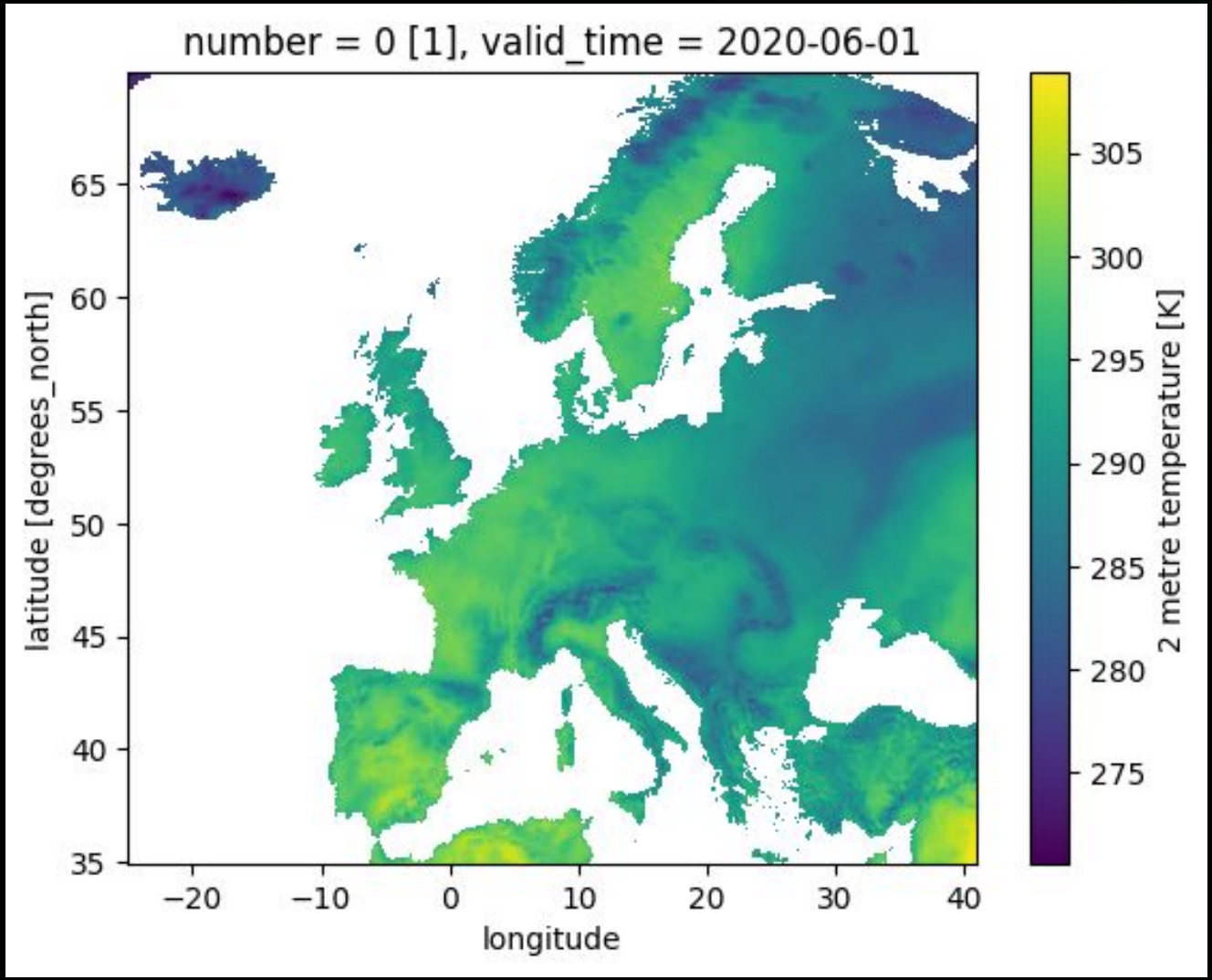
Administrative boundaries for spatial masking and ground station data for validation.

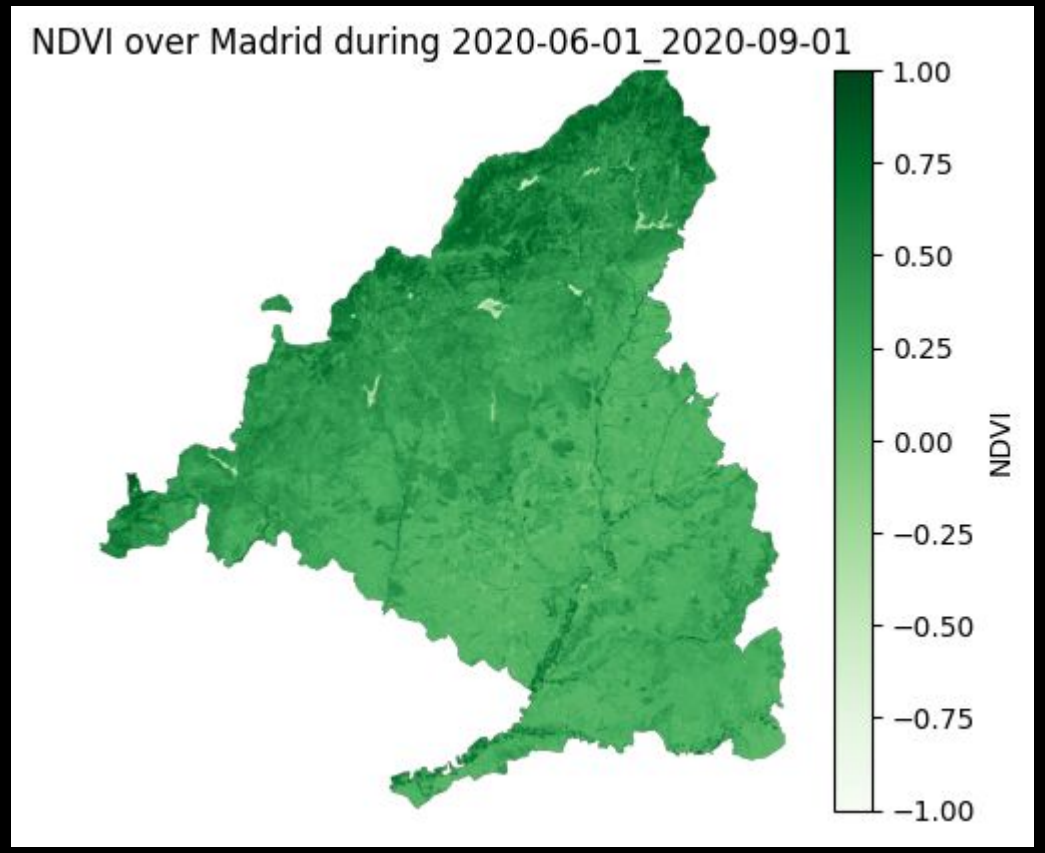
Temperature Mapping

Using ERA5-Land data, we isolated the daily maximum temperature for August 10th, 2020.

The visualization clearly shows a thermal hotspot centered over the metropolitan area of Madrid (yellowish colors), with temperatures reaching peaks significantly higher than the Sierra de Guadarrama to the northwest.

- Variable: 2m_temperature
- Metric: Daily Maximum
- Resolution: 9km x 9km





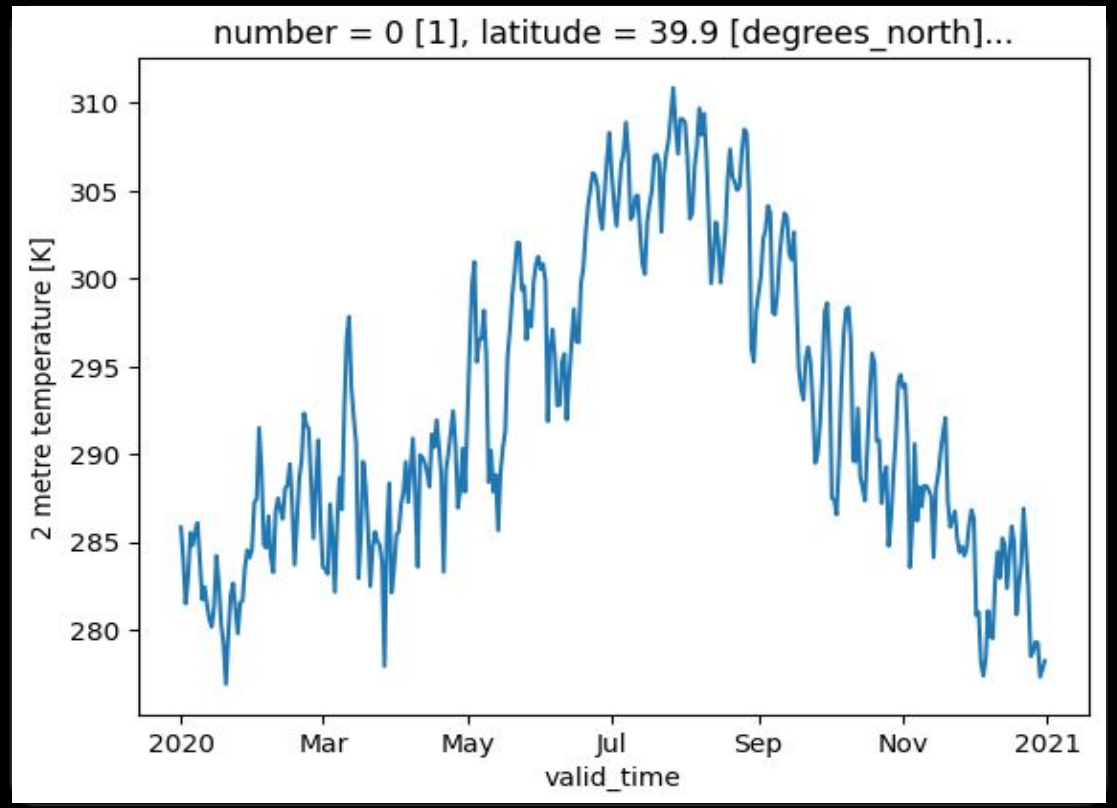
Vegetation Index (NDVI)

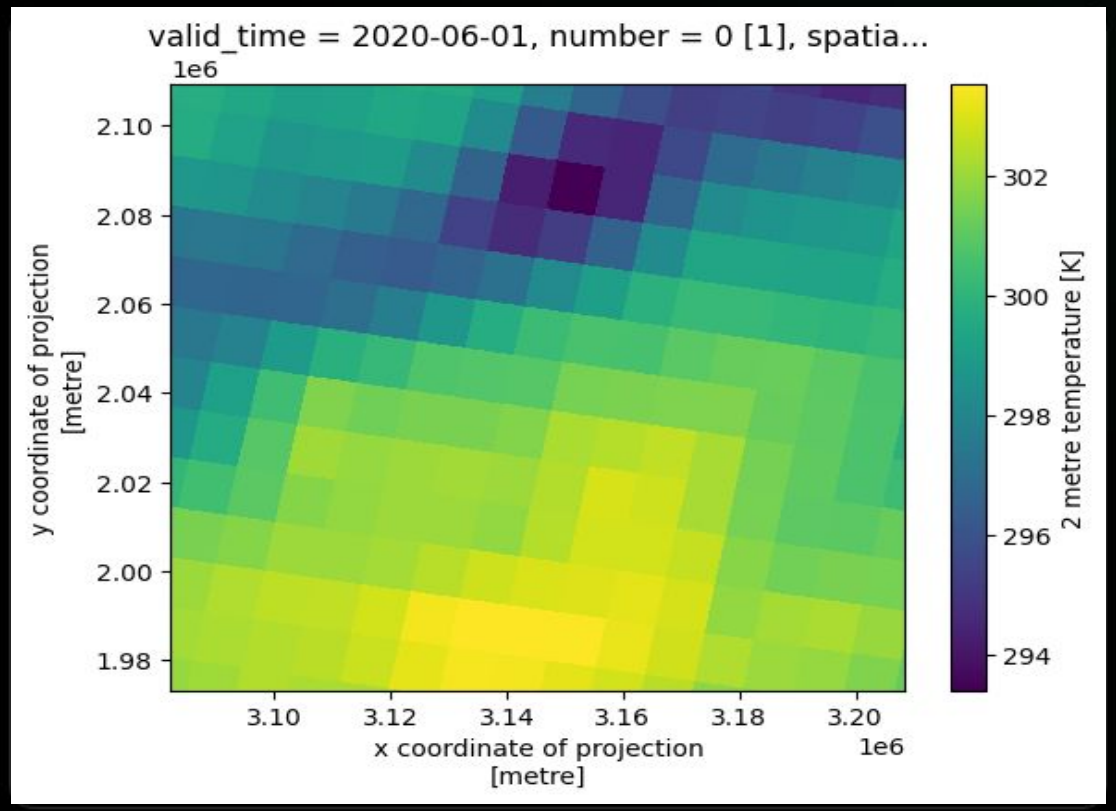
Sentinel-2 imagery allows us to quantify greenness. The Normalized Difference Vegetation Index (NDVI) ranges from -1 to +1.

In Madrid, the NDVI mean is around 0.3 indicating sparse vegetation and high impervious surfaces (concrete/asphalt), which absorb and re-radiate heat.

- Source: Sentinel-2
- Period: 2020 Quarter 3
- Scale: 0.1 to 0.8

Some results

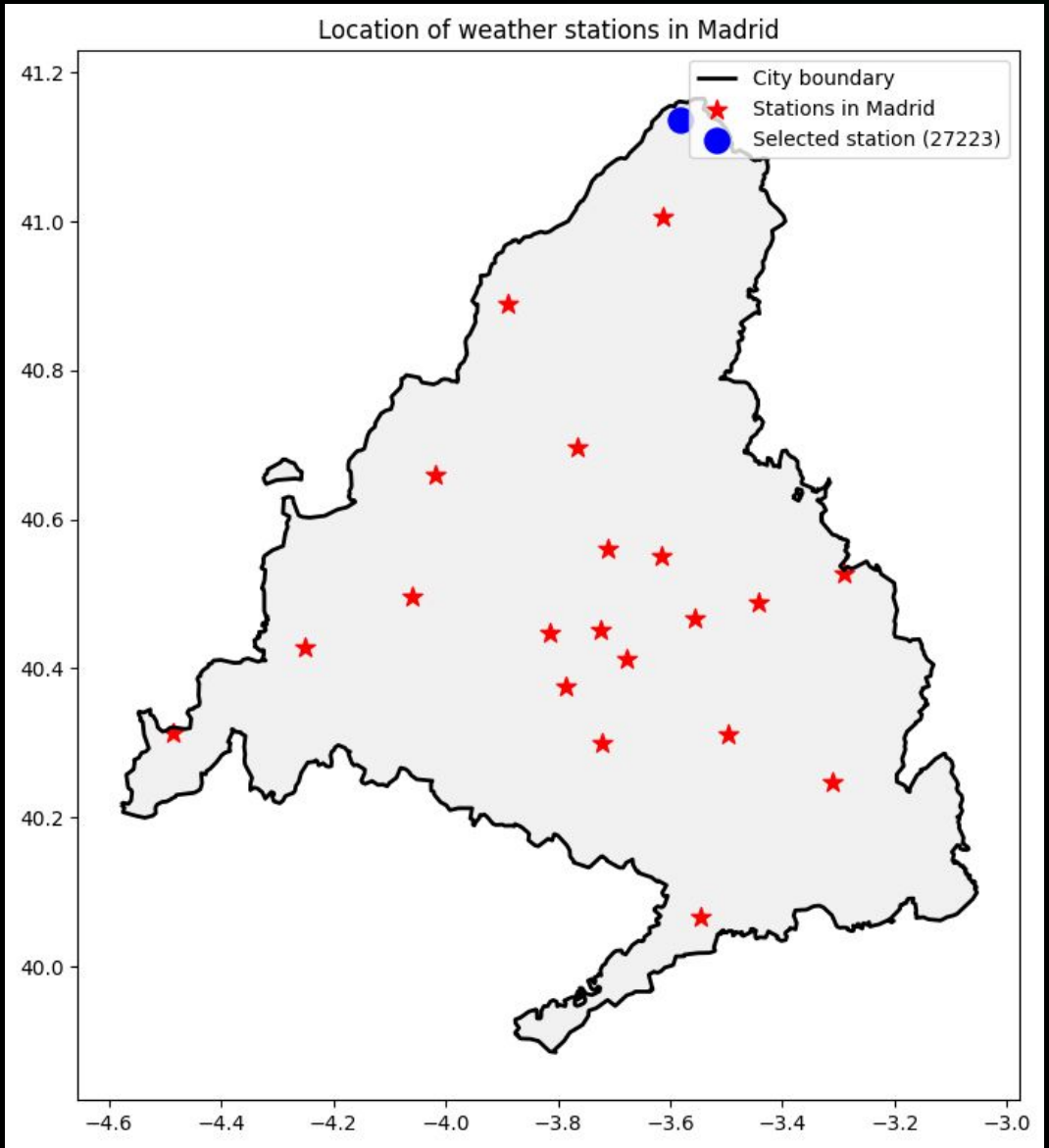




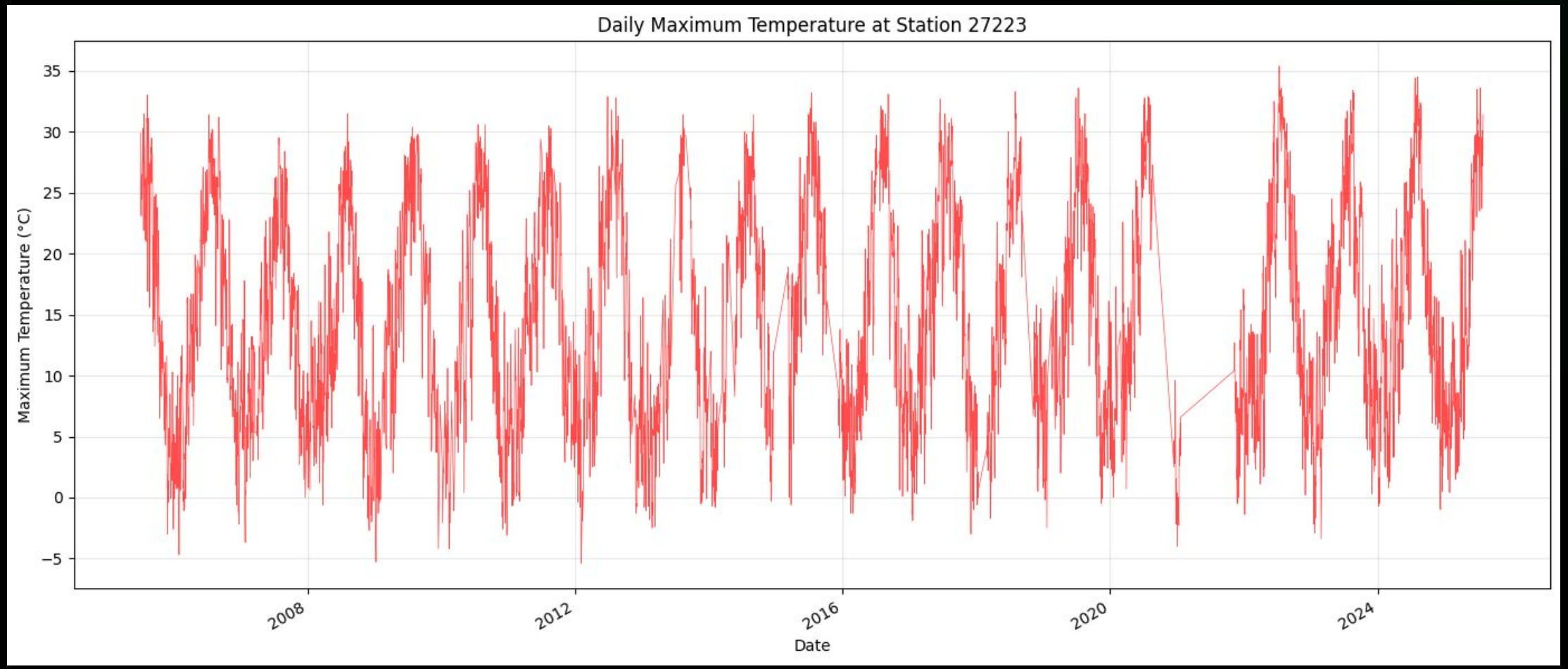
Madrid Stations

Madrid has multiple weather stations as shown in here.

For the next slide we will select **station 27223 (Somosierra)**



Summer 2020 Heatwave



Time series analysis of Station 27223 (Somosierra) confirms a sustained heatwave in August, aligning with the static map analysis.

Next Steps: Week 3

1. Quantitative Metrics

Calculate Pearson correlation coefficients between Temperature and NDVI.

2. Downscaling

Attempt to use the high-res vegetation data to "sharpen" the low-res temperature maps.

Thank You

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